



Enhancing students' critical thinking literacy in a generative AI context: Eye movement patterns of deepfake detection

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ABSTRACT

This study examined, using eye-tracking technology, the gaze patterns of guided exposure to deepfake and authentic portrait images in comparison to unguided exposure to assess their influence on the acquisition and application of critical thinking literacy among first-year Bachelor students. Employing a between-groups experimental design, we analyzed visual gaze patterns between students who participated in the experiment group ($n = 24$) and those in the control group ($n = 44$). The results revealed significant differences in visual attention patterns, with the experimental group exhibiting a longer duration and broader distribution of fixations compared to the control group. The experimental group also achieved higher accuracy in identifying deepfake portrait images and exhibited a greater propensity to classify images as GenAI-generated. Nonetheless, both groups overestimated their detection capabilities, with no significant difference in perceived performance. As GenAI technologies continue to evolve, higher education institutions must prioritize the development of the essential 21st-century literacies, integrating cognitive training with technological solutions to combat GenAI-generated misinformation. This research contributes to the fields of GenAI literacy and critical thinking education by providing objective evidence, as measured through eye-tracking, that targeted interventions can significantly alter cognitive processing strategies. Educational practice must balance the development of critical evaluation skills with fostering a realistic awareness of human limitations, thereby preparing students to become discerning consumers and responsible creators in an increasingly complex digital information landscape.

1. Introduction and aim of the study

Humanity is entering an age of Generative Artificial Intelligence (GenAI), which is rapidly permeating people's daily lives and transforming how they consume and evaluate information. This technological revolution is particularly evident in the way GenAI influences humans' perceptions of the world through visual representations, specifically images (Rapp et al., 2025; van Hees et al., 2025). Images can convey complex information quickly and succinctly, allowing people to grasp ideas faster than when reading a text (Mayer, 2024; Mayer & Sims, 1994). In addition, humans are more likely to remember information presented in images because of their brain's ability to process and store visual data more effectively (Lahner et al., 2024). However, exposing humans, especially learners, to images generated by GenAI (e.g., deepfake images) raises questions regarding the perception and understanding of their

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learning environment. Moreover, deepfake images can mimic reality in various applications and have significant ethical and security implications. Learners may be exposed to deepfake images that spread false information, leading to misconceptions about their study subjects. This highlights the urgent need to cultivate GenAI critical thinking literacy among learners - particularly by raising awareness of deepfakes - enabling them to critically evaluate, identify, and navigate GenAI-generated content while preserving the integrity of their educational experience (Roe et al., 2024).

Since eye-tracking provides objective data on the initial steps of human cognitive information processing, including visual intake, integration, and active information search patterns (Jarodzka et al., 2021; Liu & Cui, 2025), it has become a valuable tool for studying visual attention and perception across various domains. In educational settings, eye-tracking has been used to analyse learning processes and strategies (Catrysse et al., 2018; Cheng et al., 2024) and to enhance visual intelligence skills (Sindhwani et al., 2020). These measurements also offer insights into short-term brain activity, particularly through gaze patterns that reflect visual attention and decision-making (Skaramagkas et al., 2021).

In light of this, the objective of this study was to compare visual gaze patterns of guided exposure to deepfake and authentic portrait images with those of unguided exposure, to assess their influence on the acquisition and application of critical thinking literacy among higher education students.

In the next sections, we present the theoretical background that frames this study. We begin with an overview of GenAI in higher education, outlining both the opportunities and challenges it poses for learning. This is followed by a discussion of the deepfake challenge, focusing on the increasing difficulty of distinguishing between authentic and AI-generated media. The review then turns to critical thinking literacy in higher education in the context of GenAI, emphasizing the importance of analytical and evaluative skills in addressing synthetic media. Subsequently, we introduce eye-tracking as a methodological approach to studying image distinction, highlighting its relevance for investigating the cognitive processes that underlie students' ability to detect deepfakes. Finally, we present the study frame work, including the research questions and hypotheses.

2. Literature review

2.1. GenAI in higher education

Higher education institutions and their research endeavors function as dynamic ecosystems that facilitate the transfer and co-creation of knowledge through diverse pedagogical practices. The educational journey is marked by the interaction between teaching and learning, wherein faculty members serve as knowledge transmitters and facilitators of a learning environment that promotes critical thinking, problem-solving, and the development of competencies pertinent to current and future professional fields. These pedagogical practices underscore the significance of active student engagement and self-directed learning, culminating in profound and meaningful learning outcomes (Chernenko, 2020). For students, learning constitutes acquiring knowledge that augments professional capabilities in both present and future contexts (Lizzio & Wilson, 2004; Veiga Simão & Flores, 2010).

Recent research has underscored the transformative potential of GenAI for students in higher education. Tools such as ChatGPT can substantially enhance students' knowledge acquisition and learning processes across diverse subjects. They also can empower learners by granting them autonomy, which enhances their sense of control over their learning processes and bolsters their self-efficacy. Furthermore, these tools provide continuous real-time feedback on performance, potentially enriching students' learning experiences (Cotton et al., 2023; Crawford et al., 2023; Dai et al., 2023). Consequently, student engagement in the learning process is likely to increase, as is their capacity to critically evaluate information and formulate questions, thereby strengthening their critical learning skills (Cotton et al., 2023; Tajik & Tajik, 2023).

While GenAI presents considerable opportunities for higher education, it is also associated with certain concerns. Integrating GenAI into educational settings raises ethical challenges that require a thorough investigation. For example, GenAI applications facilitate content creation that is nearly indistinguishable from human-produced work. This capability may undermine the value of academic degrees and impede higher education institutions' ability to produce adequately prepared graduates in the labor market (Cotton et al., 2023; Crawford et al., 2023). Furthermore, it is found that most references to GenAI ethics in academia pertained to academic integrity issues or the misuse of information among students (Sok & Heng, 2023). These behaviors are indicative of the negative consequences associated with the use of GenAI applications in academia. One concern is the potential misuse of these applications without proper attribution, which may compromise essential academic skills, such as critical thinking, problem-solving, creativity, research abilities, and interpersonal communication (Gupta et al., 2023; Tlili et al., 2023).

Another critical issue relates to students' limited ability to recognize GenAI-generated media that is indistinguishable from human work. This difficulty exposes them to particular risks that go beyond academic integrity. This GenAI-generated media can enable advanced forms of cyberbullying, including the creation of non-consensual or defamatory content, with potentially severe consequences for students' wellbeing and future careers. Additionally, fabricated statements or visuals attributed to institutions and their figures may erode trust in universities and learning resources. The spread of false facts, manipulated visuals, or synthetic datasets also poses risks to the reliability of research and teaching materials (Roe et al., 2024). Without targeted efforts to foster digital literacy and critical thinking skills, students remain particularly vulnerable to the harms of undetected AI-generated media (Fulsher et al., 2025, pp. 1–13).

2.2. The deepfake challenge

Deepfake media, also known as Synthetic media, refers to text, images, video, and audio that has been synthesized by a machine-

learning system (Farid, 2022; Wagner & Blewer, 2019). Deepfake AI refers to the use of AI techniques to create or manipulate audio and/or visual content in a highly realistic and convincing way (Dhanaraj & Sridevi, 2024).

The ability of humans to distinguish between deepfakes and authentic images has been studied for over a decade and has intensified as GenAI has become increasingly prevalent. These studies focused on various kinds of images, starting with video images (Wagner & Blewer, 2019), food images (as part of understanding consumer preferences; Califano & Spence, 2024), and human face images as a hatch to understanding human social behaviors (Miller, Steward, et al., 2023). However, most research on deepfake detection by humans has focused on facial images (Diel et al., 2024). Faces are uniquely demanding and combine perceptual and cognitive complexities. Deepfake faces are influenced by biases in human judgment (Makowski et al., 2025), and are processed differently in the brain compared to other objects (Kanwisher et al., 1997). This makes them especially compelling for studying detection. In addition, leveraging well-documented gaze patterns in face perception (as will be presented further in the article), allows researchers to predict control group behavior more accurately, further justifying the focus on facial images (Fearington et al., 2025). The results of studies conducted by Farid and colleagues between 2007 and 2017 (Farid & Bravo, 2007, 2012; Holmes et al., 2016; Mader et al., 2017), together with the systematic review by Miller and colleagues up to 2022 (Miller, Steward, et al., 2023), revealed inconsistent findings concerning the ability to distinguish between deepfake and authentic images. Nevertheless, a recurring trend is that earlier studies tended to report a higher probability of distinguishing between the two types of images. This pattern can be explained by the presence of abnormal and easily detectable features in the AI-generated faces of that period, in contrast to the increasingly realistic and refined outputs produced by more recent GenAI systems as algorithms advanced (Lago et al., 2021).

An illustration of this shift can be found in the study by Nightingale and Farid (2022), which demonstrated that participants' performance in differentiating authentic from AI-generated faces was essentially at chance level, with accuracy hovering around 50 %. Similar findings regarding the increasing difficulty in distinguishing between authentic and AI-generated facial images are evident in a recent study by Huang and colleagues (Huang et al., 2024), reporting difficulty in distinguishing real from fake facial images: while per-participant recognition accuracy averaged around 76–80 %, the confusion matrix analyses showed that only 40 % of real images and 37 % of fake images were correctly identified across the dataset, suggesting that participants often relied on guessing strategies.

Recent studies have identified a higher tendency among participants to exhibit "AI hyperrealism", characterized by a greater inclination to perceive GenAI-generated images of faces as authentic representations of humans (<link id=bib_miller_et_al_2023b>Miller, Steward, et al., 2023; </link>; Nightingale & Farid, 2022). The AI hyperrealism phenomenon was described by Makowski and colleagues (Makowski et al., 2025). In their study, participants were requested to distinguish between GenAI and authentic images without the notion that all images were authentic. The study results presented a 54 % authentic image decision-making, emphasizing the GenAI hyperrealism phenomenon. Additionally, participants' self-reported personality examinations revealed a significant positive relationship between the attractiveness of faces and their judgment began to be authentic. The more attractive an image was perceived, the higher it was judged to be a human. This study, along with the research presented above, demonstrates that rapid advances in GenAI technology have produced higher realistic images that make it nearly impossible to distinguish between AI-generated and authentic human faces.

To address this, Bray and colleagues (Bray et al., 2023) conducted an intervention study to teach students how to identify deepfake face images. In their study, before the decision-making of face images (i.e., GenAI or real), the study group learned 10 principles for detecting GenAI face images. They were then divided into two additional groups: those receiving a reminder and those without. Although the intensity of the intervention increased the accuracy of identifying deepfake images, the overall accuracy between the groups was insignificant. A post-study survey was conducted to understand participants' reasoning "behind" their decision-making, finding that several principles were used more than others, yet not always resolving incorrect results.

These reviewed findings highlight the growing difficulty humans face in reliably detecting deepfakes and underscore the need for educational approaches that strengthen not only perceptual accuracy but also the critical thinking skills required to evaluate GenAI content in higher education contexts.

2.3. Critical thinking literacy in higher education in the context of GenAI

Critical thinking is defined as the intellectually disciplined process of actively conceptualizing, analyzing, and evaluating information gathered from observations or experiences (Paul & Elder, 2020). This definition emphasizes the active and systematic nature of critical thinking, involving skills such as conceptualization, analysis, and evaluation.

In higher education, critical thinking literacy is perceived as a crucial skill that has gained significant attention due to its importance in academic and professional success (Indrašienė et al., 2021; Liu et al., 2014). The integration of GenAI in education has further emphasized the need for critical thinking literacy. One of the well-known AI competency frameworks established by UNESCO. The framework emphasizes the importance of developing a human-centered approach to AI, fostering critical thinking, and understanding the ethical implications of AI technologies. This holistic approach to AI education goes beyond mere technical skills, aiming to cultivate students' abilities to evaluate AI systems critically, understand their societal impacts, and contribute to the development of ethical and sustainable AI (Holmes & Miao, 2023; Krause et al., 2025).

Although the rapid progression of GenAI might prompt some to assert that the human capacity to discern authentic from synthetic media visually is a 'lost cause,' this viewpoint overlooks a fundamental cognitive step that precedes and initiates deeper verification processes. Before a learner can choose to cross-reference an image, they must first adopt a mindset of active, critical scrutiny rather than passive acceptance. The ability to question an image's authenticity acts as a vital cognitive trigger. It is this disruption of passive consumption that transforms a student from a mere consumer into a critical evaluator of information, encouraging them to engage in the triangulation that the digital age demands (Pennycook et al., 2021; Vincent-Lancrin, 2023).

Several studies have highlighted the relevance and impact of critical thinking literacy on various aspects of student learning and development. For example, critical thinking dispositions and new media literacies have been found to predict students' ability to detect fake news on social media. A study involving university students revealed a positive and moderate relationship between critical thinking dispositions, new media literacies, and fake news detection abilities (Orhan, 2023). This underscores the importance of developing critical thinking skills in the context of digital literacy. Another study on UK postgraduate business school students found that while AI text generators like ChatGPT improved skills at lower levels of Bloom's taxonomy, concerns were raised about reliability, accuracy, and ethical implications (Essien et al., 2024). This highlights the complex relationship between GenAI technologies and critical thinking skills in higher education. As GenAI technologies evolve, cultivating these skills will be crucial for ensuring responsible and informed use of GenAI in education, research, and professional practice.

Higher education institutions increasingly recognize the need to incorporate critical thinking development into their curricula, with some studies suggesting using technology-supported classrooms and explicit teaching methods to foster these skills (Dori & Lavi, 2023; Falloon, 2023). The importance of critical thinking literacy extends beyond academic success, preparing students for future competencies and informed decision-making in an increasingly complex world.

Additionally, recent research shows that although explanatory interventions can improve human performance when examining critical thinking regarding AI, it may also introduce systematic biases in participants' self-perception and task confidence, particularly among those who overestimate their abilities (Hey et al., 2023). Similarly, Bray et al. (2023) found that individuals often perform at near-chance levels when distinguishing real from synthetic faces, yet still report inflated confidence in their judgments.

Taken together all the studies described, there is still a key gap in understanding the cognitive mechanisms underlying deepfake detection. Therefore, methodologies capable of capturing real-time cognitive processing, such as visual gaze patterns, are needed to inform educational interventions that prepare students to confront the perceptual and analytical challenges posed by deepfake media.

2.4. Eye-tracking method and image distinction

Eye-tracking studies involve the collection and analysis of eye movements and gaze location of participants during various visual research activities (Carter & Luke, 2020; Lai et al., 2013; Liu & Cui, 2025; Schiessl et al., 2017). It is measured by non-invasive technologies that include both stationary and mobile eye sensors, accompanied by analytical software. This technology enables the observation and measurement of individuals' visual focus, visual pattern behaviours and visual attention, providing valuable insights into human behaviour (Ball & Richardson, 2022).

The eye-tracking methodology is based on the eye-mind hypothesis, which proposes a direct link between eye movements and cognitive processes. Within this notion, eye-movements provide a window into ongoing cognitive operations, allowing researchers to observe and analyse attention allocation and information processing in real time. The assumption implies that the duration and frequency of visual focus correspond to cognitive engagement before and during the execution of a visual task (Hessels et al., 2024; Khalid et al., 2016; Schütz et al., 2011).

Within eye-tracking research frame, two key measurements are commonly used.

- (1) Fixations-brief eye pauses at a specific point in the visual field. During fixations, visual information is gathered and processed. Fixations are mainly measured temporally (measured by time), quantitatively (measured by frequency), and spatially (measured by location). The duration and frequency of fixations vary depending on the complexity of the task. For example, by examining average fixation count and duration across Areas of Interest (AOIs), researchers can identify regions that received more visual attention than others. Higher frequency and longer time of fixations mostly indicate greater cognitive engagement, reflecting either difficulty or ambiguity in task performance (Cheetham et al., 2013; Fei et al., 2023; Mahanama et al., 2022; Ye et al., 2024).
- (2) Saccades-rapid eye movements as the gaze shifts between points of visual interest. Saccades are mainly measured spatially (the amplitude and distance between two fixations), quantitatively (frequency) and temporally (time). Saccade values are dependent on the nature of the visual task. Their amplitudes tend to decrease with the increasing of task difficulty and increasing cognitive load (Mahanama et al., 2022).

Fixations and saccades function as a continuous "cycle" of visual search process. However, in eye-tracking research, the selection of specific measures for analysis typically depends on the study's design and objectives. Most studies utilize these measures as raw eye-tracking data for statistical analysis, often complemented by visual representations such as 'heat maps' and 'gaze plots' to enhance the interpretation of results (Lai et al., 2013; Lima & Ventura, 2023).

Eye-tracking is widely applied across disciplines for various purposes, including understanding cognitive processes in learning, Human-Computer Interaction (HCI), and modelling variable competition in decision-making. By analysing eye-movement patterns, researchers can, for example, identify the visual sequence, the elements that captured most attention, and how this information influences the study domain (Lai et al., 2013; Stuart, 2022).

Further to this, eye-tracking technology has emerged as a valuable tool for analysing cognitive processing. Studies have shown that eye-movements can distinguish between strategic and non-strategic learners (Catrysse et al., 2018), problem-solving (Soares et al., 2021), indicate local versus global visual attention and reaction to intervention (Butcher et al., 2025; Schreiter & Vogel, 2023). These findings highlight the growing role of eye-tracking within educational research, particularly in understanding student learning cognitive processing.

Eye-tracking has also been used in studies of facial-image decision making. Studies in this field vary in their questions and methods,

comparing human images in various forms (e.g., videos, pictures), and image sources (e.g., GenAI, Avatar). However, most of the studies use fixations as their eye-tracking measure, and within this measurement frame, most emphasize high visual attention to main centre-face features (some in the presented order): eyes, nose, and mouth while performing the decision-making tasks (Cheetham et al., 2013; Fearington et al., 2025; Gupta et al., 2020; Huang et al., 2024; Sano et al., 2024; Wöhler et al., 2021). Additionally, Huang and Colleagues (Huang et al., 2024) report a larger number of fixations and longer average fixation durations of observations when images were suspected as fake. Overlapping findings have been presented by Gupta and colleagues (Gupta et al., 2020), who found in video-based images (real, AI fake easy and AI fake difficult to identify images), a significantly higher number of fixations for real versus fake AI images, with a higher number of re-fixations on the difficult to detect AI images. The study also notes that visual attention focused only on the faces within the real images, yet in the AI fake images the visual attention also spread to the image's surroundings. Last, and contrary to Huang and Colleagues' findings, fixation duration did not differ between the three conditions.

Building on these findings, the current study used eye-tracking measures.

2.5. The present study

The objective of this study is to compare visual gaze patterns of guided exposure to deepfake and authentic portrait images in comparison to unguided exposure, in order to assess their influence on higher education students' acquisition and application of critical thinking literacy. From a theoretical perspective, this research advances understanding of human cognitive processing in deepfake facial image detection. Prior work has shown that AI systems often reproduce and exaggerate facial features, producing hyperrealistic images that are increasingly difficult to distinguish from authentic ones (Miller, Foo, et al., 2023). Given the novelty of using eye-tracking to assess cognitive shifts in response to deepfake detection training, this research is exploratory in nature. Its primary aim is to investigate initial patterns and generate preliminary insights that can inform hypotheses for future, larger-scale studies. By analyzing gaze patterns, the study examines whether guided interventions can effectively alter the cognitive processes involved in detecting deepfake facial images. Moreover, since faces require higher-order judgment that extends beyond perception into evaluation and interpretation, they are directly relevant to fostering and assessing students' critical thinking skills (Paul & Elder, 2020; Vincent-Lancrin, 2023).

From a practical standpoint, the study contributes to education and research by evaluating a specific pedagogical intervention designed to enhance critical thinking literacy in the GenAI era. By comparing gaze patterns under guided and unguided conditions, this research seeks to identify teaching strategies that can support learners in developing more deliberate and reflective approaches when engaging with potentially manipulated visual materials.

Following this, five research questions were addressed.

1. Are there differences (and to what extent) between the experiment and control groups' patterns of visual gaze attention when viewing authentic versus deepfake portrait images?
2. Are there differences (and to what extent) between the experiment and control groups in visual gaze duration when viewing authentic versus deepfake portrait images?
3. Are there differences (and to what extent) between the experiment and control groups' actual success in identifying portrait images accurately?
4. Are there similarities between the experiment and control groups in their AI hyperrealism tendency?
5. Are there differences (and to what extent) between the experiment and the control group in perceived success in identifying portrait images accurately?

Drawing on the reviewed literature and research questions, the following hypotheses were proposed to guide the present study: Eye-tracking provides a reliable method for examining cognitive processes during image-based decision-making (Fearington et al., 2025; Gupta et al., 2020; Huang et al., 2024; Sano et al., 2024; Wöhler et al., 2021). Prior research has also shown that interventions can reshape gaze behavior by directing attention toward relevant features and modifying fixation patterns (Butcher et al., 2025; Schreiter & Vogel, 2023). Based on these evidences, the following hypotheses are proposed.

H1. *Participants in the experimental group will demonstrate significantly different patterns of visual gaze attention in AOIs compared to the control group.*

H2. *Participants in the experimental group will demonstrate significant differences in the duration of eye gaze compared to the control group.*

Prior studies highlight the importance of fostering critical thinking and digital literacy to help students detect manipulated media, with evidence that these skills predict the ability to identify fake news and synthetic content (Holmes & Miao, 2023; Krause et al., 2025; Orhan, 2023). Therefore, the following hypothesis is proposed.

H3. *Participants in the experimental group will be significantly more accurate in identifying deepfake and authentic portrait images compared to the control group.*

The rapid advancement of GenAI has led to increasingly realistic human portrait images, producing the phenomenon of AI hyperrealism, where participants often misclassify synthetic faces as authentic (Miller, Foo, et al., 2023; Nightingale & Farid, 2022; Makowski et al., 2025). In light of these findings, the following hypothesis is proposed.

H4. *No statistically significant differences in AI hyperrealism tendency will be observed between the experimental and control groups.*

Research also shows that explanatory interventions may improve detection performance but introduce systematic biases in self-perception and task confidence, particularly overestimating their own abilities (Hey et al., 2023; Bray et al., 2023). Therefore, the following hypothesis is proposed.

H5. *The experimental group will outperform the control group in the perceived identification of deepfake vs. authentic images.*

In according to these theoretical research frames, we designed the study.

3. Method

The study, approved by the Ethical Committee at the Holon Institute of Technology, employs a between-group design, incorporating both an experimental group and a control group. A between-group design (also called a between-subjects design) is an experimental research design in which participants are assigned randomly to separate groups. One group received an intervention (experimental) and the other did not (control). The outcomes of these independent groups are then compared to determine the effect of the manipulated variable (Gravetter & Forzano, 2018).

3.1. Participants

The study included a total of 76 participants (65 female and 18 male). Most (90 %) of the participants were first-year Bachelor's degree students between the ages of 18–32 ($M = 24$, $sd = 2.8$). All participants were recruited from the same higher education institution. 60 % reported they did not use at all GenAI, 32 % used it rarely and only 7 % used GenAI several times a week.

Participants were assigned to three groups: a pretesting group ($n = 8$, feasibility study; see Appendix A), an experimental group ($n = 24$, received intervention), and a control group ($n = 44$, no intervention). The experimental and control groups were drawn from two intact first-year "Visual Communication" classes, resulting in unequal group sizes due to natural enrolment differences. All participants met standard eye-tracking criteria (normal or corrected-to-normal vision; no colour blindness).

3.2. Measures

3.2.1. Deepfake detection task

The purpose of the task was to evaluate participants' visual behaviour and decision making regarding the nature of 12 human portrait images (i.e., "12 stimuli") presented on a computer screen and recorded using an eye-tracking device. Participants were requested to decide whether the image presented was conducted by GenAI (i.e., deepfake) or an authentic human image. The task was conducted linearly. Images were presented sequentially following a written explanation and a one-trial practice (to ensure instruction comprehension). The stimuli set was equally distributed for image source (6 AI-generated, 6 authentic) and gender representation (3-woman profiles per source category). For each image, participants were asked to decide whether the portrait (see example in Fig. 1a and b) is authentic or generated by AI and accordingly click the appropriate response button. No time constraints were imposed. The AI-generated portrait images were created using Gen AI this person does not exist reservoir, and the authentic images were from freepik. To maintain consistency, all 12 portraits—authentic and AI-generated, were presented with identical framing, head size, and positioning. Within this controlled framing, the images vary in the synthetic set, in typical GAN artifacts (asymmetries, background distortions, blurred accessories, irregular details).

3.2.2. Post-Hoc survey

The survey consisted of two sections: (1) A "Self-prediction of performance" question (Vuurro & Metcalfe, 2022), also known as a

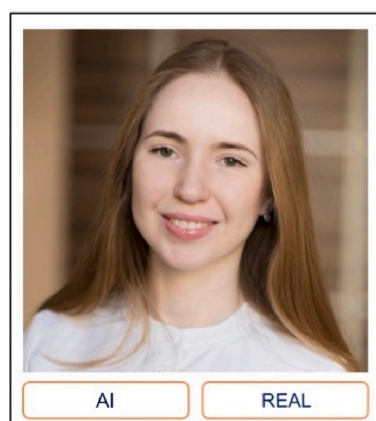


Fig. 1a. An example of a stimulus portrait of an authentic human image.

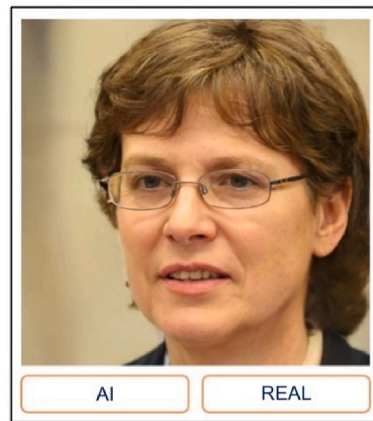


Fig. 1b. An example of a stimulus portrait generated by AI (deepfake).

confidence task (Lago et al., 2021), in which participants were asked to estimate (from 0 to 100 %) how well they distinguished in the **detection task**. (2) Demographic and AI image experience - included demographic information (e.g., age, educational affiliation) and a question regarding the experience in using and creating images by GenAI if at all.

3.2.3. Critical thinking intervention program

The intervention program was conducted *only* in the experiment group ($n = 24$) and prior to the study. Its purpose was to teach and practice the main strategies for evaluating and detecting deepfake AI versus authentic images. The program was designed and taught in class by a faculty member as part of the “Visual Communication” classes. Image analysis referred to two levels: (1) denotative level-what do we see in the image? (2) connotative level-what message does the image convey and what interpretation can be given to the image? Additionally, components of the importance of the context (in which the image is displayed, its background etc.), and up to date strategies were studied (Lago et al., 2021; Nightingale & Farid, 2022). Training included 18 images (deepfake by AI and authentic

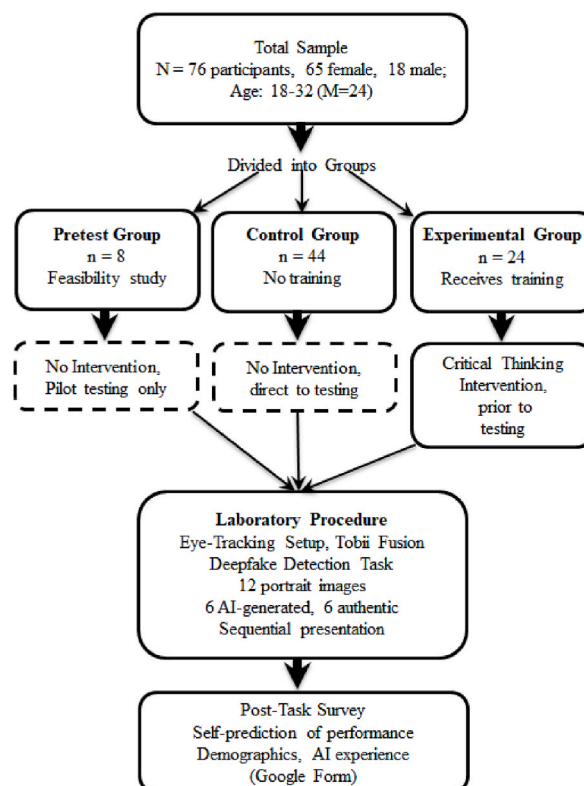


Fig. 2. The study flowchart.

human image) and students used the identification marks they learned to identify images correctly.

3.3. Procedure

The study was conducted at the “Eye-Tracking Lab” of a higher education institution. For the purposes of maintaining anonymity participants were coded by subject numbers.

During the task, eye movements were recorded using the eye-tracker “Tobii Fusion” (120 Hz sampling rate), connected to a 32-inch LCD monitor (1920 × 1080 resolution). Participants were seated 60–70 cm from the display, with stimuli presented in full-screen mode.

The procedure followed a standardized protocol (Fig. 2): First, a participant registration and anonymous coding, followed by an eye-tracking calibration (of nine-point). Next, participants read the instructions presented on screen, completed the trail and conducted the “**Deep Fake Detection Task**” (which was recorded by the eye-tracker). At the end, participants answered the “post-huck survey” on a google form.

4. Data coding and analysis

Using “Tobii Pro-Lab” software, each stimulus (human portrait images) was defined with a number from 1 to 12 in according to the order of presentation in the task. Within each stimulus, see also Fig. 3, two main Areas Of Interest (AOI’s) were defined: (1) Facial region AOI: eyes, nose, mouth (2) Background AOI: areas without facial region. These AOIs are well established in eye-tracking research on face processing (Fearington et al., 2025; Gupta et al., 2020; Huang et al., 2024; Sano et al., 2024; Wöhler et al., 2021). AOI detection is a common approach for analysing eye-tracking data, particularly fixation patterns (Vehlen et al., 2022). In this study, we focused on two measures: fixation count, which indicates attentional allocation to areas of interest (Catrysse et al., 2018; Cheetham et al., 2013), and fixation duration, which reflects depth of processing and cognitive effort (Sano et al., 2024). These measures are widely used to evaluate how interventions shape gaze behaviour in learning and decision-making tasks (Schreiter & Vogel, 2023; Soares et al., 2021)

Following the AOI coding, eye-tracking data was extracted, and a cleaning procedure was conducted using Python (Appendix B). Visualization through heat maps provided an initial overview of gaze patterns (H1), which was then followed by statistical analyses (H2–H5) conducted with IBM SPSS Statistics 28.0.

In order to understand the relationship between fixation count and fixation duration, a preliminary analysis was conducted using a Pearson correlation test. Results revealed a strong positive correlation (Pearson’s $r = 0.972$), indicating that participants who exhibited a higher number of fixations on a stimulus also tended to have longer total fixation durations. Based on this strong association, total fixation duration was selected as the primary measure for subsequent analyses, as presented in the results.

5. Results

H1: Participants in the experimental group will demonstrate significantly different patterns of visual gaze attention in AOIs compared to the control group.

A visualization comparison was conducted using eye gaze heatmaps of fixation duration for the experiment and control groups across the different facial images. To visualize the analysis process, we present the results of two representative images: one of a man and one of a woman (Fig. 4). In this figure, heatmap (A) represents the eye-tracking data of the control group for the woman image, while heatmap (B) corresponds to the experiment group for the same image. Similarly, heatmap (C) displays the eye-tracking data for the control group with the man image, and heatmap (D) represents the test group for the same image.

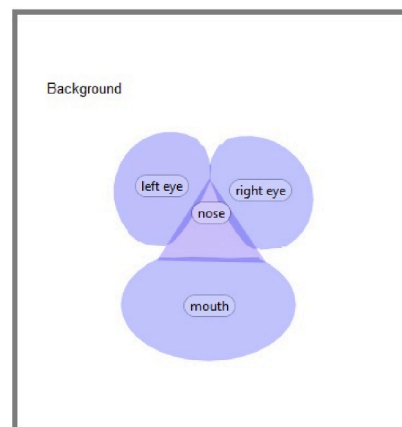


Fig. 3. AOI outline for each stimulus.

Overall, a comparison between the experiment and control groups reveals a notable difference in total fixation duration distribution. In both pairs of images, the experiment group (B, D) exhibited a wider fixation in areas outside the eyes, nose, and mouth, specifically in the "background" or "facial periphery" area of interest regions. This trend is clearly observed in image B, which features a woman's face, where the experiment group exhibited increased fixations around the ears and hair compared to the control group (A). A similar pattern is evident in image D, featuring a man's face, where the experiment group showed more fixations in these areas compared to the control group (C).

Similar results were obtained for the other images in the study. Based on our results, [H1](#) is accepted.

H2: *Participants in the experimental group will demonstrate significant differences in the duration of eye gaze compared to the control group.*

Phase 1: Quantitative analysis of two representative images

A boxplot illustrating the distribution of total fixation duration across different areas of interest was generated. [Fig. 5](#) presents the distribution of total fixation duration for each AOI (area of interest). Panels A and B display the distributions for the representative images: the woman's and man's faces, respectively. An independent sample tests for fixation duration indicated a statistically significant difference in total fixation duration between the experiment and control groups. For the woman face, the total fixation duration (average $\pm$ standard deviation) in the control group was 4425 ± 4653 ms, while in the test group, it was 9526 ± 5839 ms (t -test, $p < .01$). Similarly, for the man face, the total fixation duration in the control group was 3802 ± 3400 , whereas in the test group, it was 8172 ± 5250 ms (t -test, $p < .01$).

Following these results, we examined which AOI's contributed to the observed differences. It can be seen that in both graphs ([Fig. 5](#)), a significant difference in total fixation duration is seen within the background/facial peripheral areas. For the woman's face, the average total fixation duration within the background in the control group was $1269 \pm 1,933$ ms, while in the experiment group, it was $4496 \pm 4,218$ ms (t -test, $p < .01$). Likewise, for the man face, the average total fixation duration within the background in the control group was $1195 \pm 1,668$, whereas in the experiment group, it was $4012 \pm 3,369$ ms (t -test, $p < .01$).

We chose to focus on the background, as it was found to be the most significant among all areas of interest. The background showed the largest significant difference between the experimental and control groups, appearing in 10 out of the 12 experiments. For example, when analyzing the eyes as an area of interest, it was found that in the woman face image, the total fixation duration (in the eye) in the control group was $1894 \pm 2,101$ ms, while in the experiment group, it was 2803 ± 1615 ms (t -test, $p = .052$). Similarly, for the man face image, the total fixation duration (in the eye) in the control group was $1703 \pm 1,556$ ms, whereas in the test group, it was $2049 \pm 1,557$ ms (t -test, $p = .385$).

For the sake of simplicity, all the results are presented both graphically in [Fig. 5](#) and in a table ([Table 1A](#), [Table 1B](#))

One might argue that the observed difference is a result of an extension in individual total fixation duration rather than an increase in the number of fixations. This hypothesis was tested, and the results showed no significant difference in the fixation duration between the groups. The average fixation duration in the experimental group was 289 ± 180 ms, while in the control group it was 290 ± 148 ms (t -test, $p = .90$) (see details in [Appendix C](#)).

Phase 2: Quantitative analysis of all images

A boxplot that visualizes the total fixation duration within the background AOI's across all images for both the experiment and control groups was created ([Fig. 6](#)). The mean values, standard deviations, and p-values for all images (stimuli) are presented in [Table 2](#). Both the table and the graph indicate that, except for the last stimuli, there is a significant difference ($pval < 0.05$) between the control and experiment groups in the total fixation duration within the background AOI's.

Using the binomial model, the probability of obtaining randomly a result where all outcomes are statistically significant except for one is 10^{-14} .

The binomial model is a complex framework, yet it allows for differences that may arise between the various images and measures. In order to simplify the analysis, and have a wider view, we summarized the results of the experimental group and the control group separately. The comparison between them shows that even when aggregating all results, a significant difference was found between

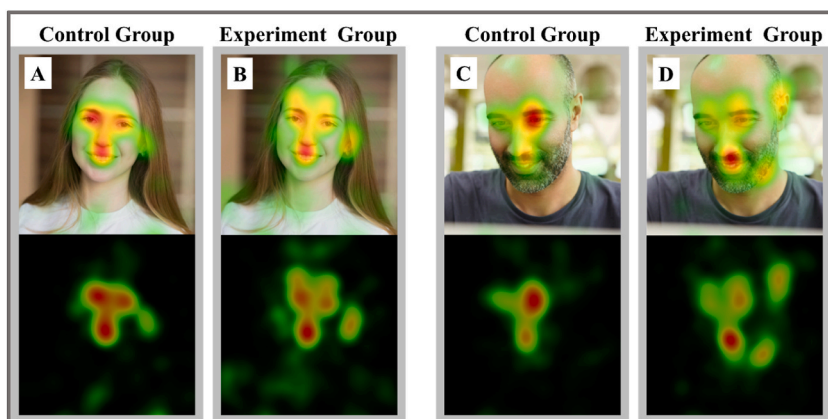


Fig. 4. Eye-tracking heatmaps for two representative images across the two groups-experiment and control.

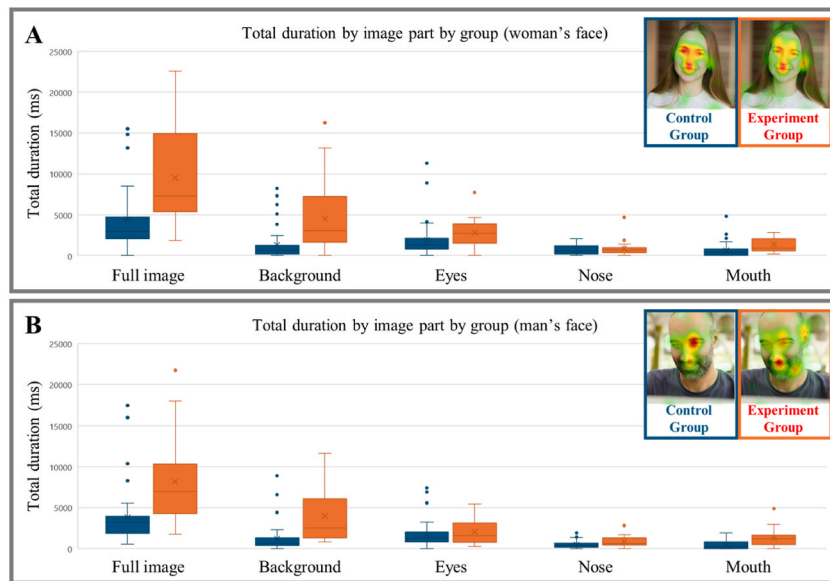


Fig. 5. Distribution of total fixation duration across different AOI's for woman face image (A), and man face image (B).

Table 1

Average and standard deviation of total fixation duration across different AOI's for woman face image (A), and man face image (B).

A			
Stimulus	Control Average $\pm$ SD	Experiment Average $\pm$ SD	T. test
Full image	4425 $\pm$ 4653	9526 $\pm$ 5839	-3.688***
Background	1269 $\pm$ 1933	4496 $\pm$ 4218	-3.550**
Eyes	1894 $\pm$ 2101	2803 $\pm$ 1615	-1.987
Nose	665 $\pm$ 558	859 $\pm$ 956	-0.914
Mouth	597 $\pm$ 894	1368 $\pm$ 1095	-2.952**
B			
Stimulus	Control Average $\pm$ SD	Experiment Average $\pm$ SD	T. test
Full image	3802 $\pm$ 3400	8172 $\pm$ 5250	-3.678***
Background	1195 $\pm$ 1668	4012 $\pm$ 3369	-3.847***
Eyes	1703 $\pm$ 1556	2049 $\pm$ 1557	-0.877
Nose	465 $\pm$ 442	835 $\pm$ 662	-2.461*
Mouth	440 $\pm$ 572	1275 $\pm$ 1117	-3.424**

* $p < .05$, ** $p < .01$, *** $p < .001$.

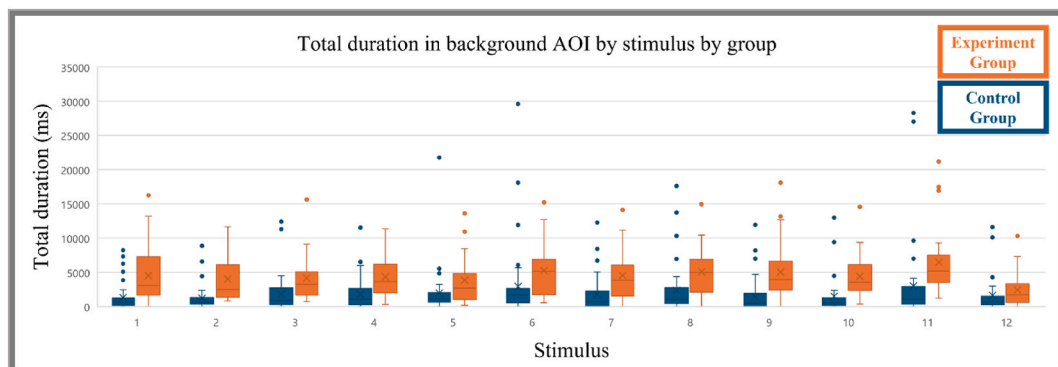


Fig. 6. Total fixation duration within the background AOI's across all images for both groups.

Table 2

Fixation duration within the background areas across stimuli, for both the experiment and control.

Stimulus	Control Average $\pm$ SD	Experiment Average $\pm$ SD	T. test
Image 1	1269 $\pm$ 1934	4496 $\pm$ 4219	-3.55**
Image 2	1195 $\pm$ 1668	4012 $\pm$ 3369	-3.85***
Image 3	1784 $\pm$ 2576	4142 $\pm$ 3502	-2.9 **
Image 4	1787 $\pm$ 2313	4316 $\pm$ 3094	-3.51**
Image 5	1961 $\pm$ 3336	3830 $\pm$ 3503	-2.14*
Image 6	2937.4 $\pm$ 5227	5235 $\pm$ 3920	-2.05*
Image 7	1761.43 $\pm$ 2641	4527 $\pm$ 4127	-9.7**
Image 8	2361.9 $\pm$ 3562	5069 $\pm$ 3442	-3.06**
Image 9	1422.0 $\pm$ 2456	5022 $\pm$ 4360	-3.73***
Image 10	1492.25 $\pm$ 2603	4396 $\pm$ 3356	-3.68***
Image 11	2981 $\pm$ 5896	6446 $\pm$ 5179	-2.41*
Image 12	1622 $\pm$ 2767	2430 $\pm$ 2558	-1.21
All	1881 $\pm$ 3328	4494 $\pm$ 3815	-10.17***

* $p < .05$, ** $p < .01$, *** $p < .001$.

the experimental and control groups, with (see Table 2).

This result demonstrates that there is a significant difference between the experiment and control groups regarding the total duration of background fixations. Based on our results, H2 is partially accepted.

H3: *Participants in the experimental group will be significantly more accurate in identifying deepfake and authentic portrait images compared to the control group.*

To address the second hypothesis, the differences between the groups were initially analyzed for each image individually, followed by an assessment of the overall score.

For the initial analysis, Chi-square tests were conducted to examine the relationships between group membership (experiment vs. control) and response patterns (correct vs. incorrect answers) for each image. As presented in Table 3, A significant association was found in 8 (out of 12) images between group membership and the correctness of responses. However, the distribution of correct and incorrect answers between the two groups differs across images, and both groups demonstrated low accuracy in all images.

Furthermore, a Chi-square test was conducted to examine the relationship between total response patterns and group membership (experimental vs. control group). A significant association was identified between the correctness of responses and group membership ($\chi^2 = 23.54$, $p < .01$). As illustrated in Table 3, within the 'Total response' row, the experimental group exhibited a higher proportion of correct responses (151 correct vs. 137 incorrect) compared to the control group, which showed a higher proportion of incorrect responses (340 incorrect vs. 188 correct). Based on our results, H3 can be accepted.

H4: *No statistically significant differences in AI hyperrealism tendency will be observed between the experimental and control groups.*

A Chi-square test was conducted to examine the relationship between group membership (experimental vs. control) and the detection of images as deepfakes. The analysis revealed a highly significant association between group membership and deepfake detection ($\chi^2 = 35.18$, $p < .001$). Participants in the experimental group demonstrated a greater propensity to classify images as deepfake (185 AI classifications vs. 103 non-AI classifications), whereas the control group exhibited the opposite tendency (325 non-AI classifications vs. 203 AI classifications). Based on our results, H4 is rejected.

H5: *The experimental group will outperform the control group in the perceived identification of deepfake vs. authentic images.*

Upon completing the detection task, participants were asked to estimate their percentage of correct responses, with a range of 0 %–100 %. In this section, we compared participants' self-assessed estimates with their actual scores on the decision-making task, with scores evaluated as percentages. Fig. 7 illustrates the average percentage ranking of the two measures within each group. A Mann-

Table 3

Distribution of correct and incorrect responses for each stimulus across the two experimental groups.

Stimulus	Correct answer		Incorrect answer		Chi-square (df = 1)
	Experiment group (N = 24)	Control group (N = 44)	Experiment group (N = 24)	Control group (N = 44)	
Image 1, authentic	9	34	15	10	10.57***
Image 2, authentic	6	18	18	26	1.72
Image 3, GenAI	17	4	7	40	27.73***
Image 4, authentic	7	14	17	25	1.292
Image 5, GenAI	19	12	5	32	16.86***
Image 6, GenAI	19	12	5	32	16.86***
Image 7, authentic	8	19	16	25	0.63
Image 8, GenAI	17	16	7	28	7.39**
Image 9, authentic	10	6	14	38	6.78**
Image 10, authentic	15	28	9	16	0.009
Image 11, GenAI	13	12	11	32	4.83*
Image 12, GenAI	11	8	13	36	5.89*
Total response	151	188	137	340	

* $p < .05$, ** $p < .01$, *** $p < .001$.

Whitney *U* test indicated no significant difference between groups in terms of participants' perceived identification of deepfake versus authentic images. Both groups assessed their performance at an average of 68 % (experimental group) and 66 % (control group) correct responses. Based on our results, H5 is rejected.

6. Discussion

This exploratory study examined, using eye-tracking technology, the gaze patterns of guided exposure to deepfake and authentic portrait images in comparison to unguided exposure to assess their influence on the acquisition and application of critical thinking literacy among first-year Bachelor students.

Eye-tracking represents a central methodological contribution of this study, as it provides objective indicators of participants' cognitive processing during the intervention. Prior research has shown that eye-movements can distinguish learning strategies, support analysis of problem-solving, and capture local versus global visual attention (Catrysse et al., 2018; Schreiter & Vogel, 2023; Soares et al., 2021). By applying this method to the context of deepfake detection, the study not only advances understanding of how guided versus unguided exposure shapes gaze behaviour but also demonstrates the broader potential of eye-tracking for investigating how educational interventions influence students' critical thinking processes.

The results, while preliminary, show a significant difference in visual attention patterns between the experimental and control groups. These initial findings suggest that targeted training can alter the cognitive processing strategies students employ when evaluating GenAI-generated content. Specifically, the experimental group, which received critical thinking intervention, exhibited a wider distribution of fixations compared to the control group (H1 accepted, H2 partially accepted).

As in previous studies (Cheetham et al., 2013; Gupta et al., 2020; Huang et al., 2024; Sano et al., 2024; Wöhler et al., 2021), both groups spend similar total fixation durations across the main facial features of eye, nose and mouth. Focusing on these features while examining a face is mainly explained by psychological and social aspects, since these features are critical for conveying emotional expressions, social signals, and identity recognition (Porras & Louwerse, 2025).

However, the finding that the experiment group significantly expanded their visual attention to peripheral areas (higher total fixation duration of background AOI's), may indicate a shift from intuitive visual examination strategy to a more comprehensive evaluation approach, emphasizing the interventions impact. This finding is particularly significant given the challenges identified in previous deepfake detection research (Gupta et al., 2020; Huang et al., 2024). Huang et al. (2024) found that an image's background plays a key role in how viewers judge its source. Yet, most visual attention typically focuses on the central facial features - the eyes, mouth, and nose (Fearington et al., 2025). Building on this, the current study suggests that the intervention enhanced students' critical thinking literacy and encouraged them to broaden their visual attention to image areas that support more accurate deepfake detection.

The significantly better overall detection accuracy demonstrated by the experiment group in comparison to the control group (H3 accepted) further supports the intervention's positive impact on students' practical skills in detecting deepfakes. Nevertheless, it is crucial to note that both groups still exhibited relatively low accuracy for individual images. This outcome aligns with the literature, which consistently reports the inherent difficulty humans face in distinguishing between GenAI-generated content and authentic media, especially as GenAI technology rapidly advances (Huang et al., 2024; Nightingale & Farid, 2022). While the intervention improved overall accuracy, the persistent challenge at the individual image level underscores the sophistication of current deepfake algorithms. This suggests that even with explicit guidance, the visual cues for detection can be subtle, inconsistent, or even absent in highly convincing deepfakes. The results align with those of Bray et al. (2023), who found that while an intervention increases accuracy, the overall improvement may not always be substantial enough to overcome the fundamental difficulty posed by highly realistic synthetic media. Diel et al. (2024) suggested combining several intervention strategies, such as feedback training and raising awareness, to improve detection. The implication for education is that, while training is beneficial, it must be continuous, diverse and adaptive to the evolving visual capabilities of GenAI.

The findings revealed that the control group tended to classify the images as authentic compared to the experimental group. The control group's tendency to classify more images as authentic, even when they were deepfakes, aligns with the documented AI hyperrealism bias., e.g., a higher tendency to choose GenAI images of faces as authentic humans (Nightingale & Farid, 2022). Participants

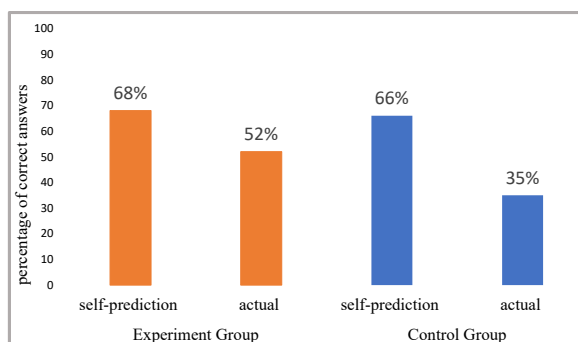


Fig. 7. The average percent of self-predict versus actual correct answers by groups.

in the experimental group demonstrated a greater propensity to classify images as deepfakes compared to the control group (H4 rejected). This result is consistent with findings by [Chen et al. \(2025\)](#) and [Rehman et al. \(2025\)](#), who also observed that AI detection training increases participants' tendency to identify images as fake. These findings carry critical implications for educational practice. While training can improve visual sampling strategies and help mitigate biases such as hyperrealism, it may also introduce a new bias toward over-identifying deepfake. To translate improved information-gathering into more accurate judgments, additional pedagogical elements are needed.

By providing specific principles and strategies for deepfake detection, the intervention equipped students with cognitive tools to critically scrutinize images rather than defaulting to an assumption of authenticity. This is a critical outcome for critical thinking literacy in a GenAI-saturated information environment, as it demonstrates the potential to mitigate inherent biases that can lead to misinterpretations of digital content.

Contrary to the hypothesis, there was no significant difference between the experimental and control groups in their perceived success of identification (H5 rejected). Both groups similarly overestimated their performance, with self-assessed estimates significantly higher than their actual accuracy. This finding aligns with the existing literature ([Köbis et al., 2021](#)), which suggests that individuals often struggle to accurately gauge their performance on complex tasks, especially when guessing is involved ([Vuorre & Metcalfe, 2022](#)). This persistent overestimation, even in the experiment group, suggests that while the intervention improved actual detection skills and altered visual gaze patterns, it did not necessarily enhance metacognitive awareness of those improvements. It is possible that the cognitive load associated with the deepfake detection task made it challenging for participants to accurately monitor their performance. Another explanation, also suggested by [Köbis et al. \(2021\)](#), is the Dunning–Kruger effect ([Kruger & Dunning, 1999](#)). According to this effect, individuals with limited competence are more likely to misjudge their performance. This may indicate that participants in the control group were overconfident, whereas participants in the experimental group had a more realistic, though still inflated, self-assessments. Consequently, both groups reported similar levels of perceived success, despite actual differences in accuracy. This highlights a crucial pedagogical challenge: not only teaching students how to evaluate GenAI content critically but also helping them develop a realistic understanding of their capabilities and limitations in doing so. This metacognitive gap could lead to overconfidence, potentially making students more vulnerable to sophisticated deepfakes in real-world scenarios.

In sum, this study contributes to the growing body of knowledge on GenAI literacy and critical thinking pedagogy, paving the way for educational practices that empower students to become discerning consumers and responsible creators in a GenAI-driven world. Using eye-tracking technology in this examination contributes to educational technology research by demonstrating how such measures can reveal the mechanisms through which educational interventions modify cognitive processing. The use of eye-tracking technology proved to be valuable in revealing the underlying cognitive processes, offering a deeper understanding of how students engage with visual information when attempting to detect deepfakes. The integration of eye-tracking into pedagogical evaluation provides evidence-based foundations for educational interventions that move beyond declarative knowledge about GenAI into procedural and strategic knowledge essential for higher education students.

Framing the task of visual discernment as futile risks overlooking a fundamental cognitive step that precedes and initiates deeper verification processes. While the rapid progression of GenAI might lead some to argue that the human capacity to distinguish authentic from synthetic media is a “lost cause,” this perspective neglects the essential role of visual analysis within the broader framework of critical evaluation. The primary goal of pedagogical interventions, such as the one examined in this study, is not to create infallible human detectors but to foster this initial stage of cognitive engagement. By requiring individuals to actively assess the authenticity of images, such interventions stimulate cognitive and metacognitive processes that are essential for the development of critical thinking skills.

By training students to move from passive consumption to active scrutiny, we equip them with an essential foundation for recognizing and questioning misinformation. This study uses eye-tracking to objectively measure whether such an intervention can shift the underlying cognitive processing strategies that are essential for navigating the complex GenAI landscape.

As GenAI technologies continue to evolve, higher education institutions must prioritize the development of the essential 21st-century literacies, integrating cognitive training with technological solutions to address GenAI-generated misinformation. Educational practice must balance the development of critical evaluation skills with fostering a realistic awareness of human limitations, thereby preparing students to become discerning consumers and responsible creators in an increasingly complex digital information landscape.

7. Limitations and future studies

Several limitations should be acknowledged, informing future research directions. It is essential to interpret the findings of this study in light of its exploratory design and its associated constraints. The primary limitation is the small sample size ($n = 24$ in the experimental group), which affects the statistical power of our analyses and limits the generalizability of the results. While the between-groups design enhances internal validity, we acknowledge that unmeasured factors could have influenced the outcomes. Therefore, the value of this research lies in its novel use of eye-tracking to reveal cognitive patterns, providing a foundation upon which future, more robust confirmatory studies can be built.

The study focused on portrait images in a controlled academic setting, raising questions about the generalizability of its findings to other deepfake formats and real-world application contexts. At the same time, unmeasured factors, such as prior exposure to GenAI, baseline analytical thinking, or cognitive style, may have influenced the outcomes.

Furthermore, this study's analysis focused on the overall differences between the experimental and control groups. As shown in the results, the distribution of correct and incorrect answers between the two groups differs across images, and both groups demonstrate low accuracy in all images. However, we acknowledge that several participants in the control group were able to identify several

deepfake images without any formal training. It is possible that they acted intuitively and made decisions mainly based on guessing.

Future research should investigate broader and longer-term interventions, accompanied by follow-up assessments, to evaluate the retention and transfer of skills to real-world contexts. Investigating personalized or adaptive training approaches, collaborative learning methods, and integrating GenAI-assisted detection tools with human judgment represents promising directions. Additionally, research examining individual differences in deepfake detection capabilities could inform more targeted educational interventions.

The persistence of detection difficulties and metacognitive overconfidence suggests that comprehensive digital literacy education requires multi-faceted approaches. Future interventions could incorporate post-experiment interviews and include: (1) explicit training in recognizing uncertainty and cultivating appropriate skepticism, (2) guided practice with feedback to enhance metacognitive accuracy, and (3) the development of broader contextual analysis skills that extend beyond visual features to encompass source credibility and situational factors. In addition, the use of concurrent think-aloud protocols may provide a clearer understanding of participants' judgment processes.

CRedit authorship contribution statement

Hayley Weigelt: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Elad Segev:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Gila Kurtz:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Omri Kahana:** Visualization, Software, Investigation, Data curation. **Nohar Raz Fogel:** Resources.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this paper the authors used “Scispace” and “Elicit”, in order to search for relevant articles which were than located in academic databases and read by the authors. After using these tools, the authors reviewed and edited the content as needed and are full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Appendix A - Findings from the feasibility study, August 2024

The feasibility study was conducted in order to ensure the experiment flow. In this manner, we mainly examined the degree of instruction clearance, stimuli size (i.e., pictures size for suitable subject inquiry) and technical aspects regarding eye-tracking collection.

The feasibility study was held during August 2024. It included seven participants (5 female, 2 male). The study durations were as expected: approximately 10–15 min (including calibration and general instructions). Picture decisions making on the screen ranking between 2:30 to 5 min (Mean = 3.65 min, SD = 0.95).

Following response of the first two participants, instructions were shortened (for better comprehension) and pictures size slightly reduced. The next six participants understood the instructions and reported the pictures were clear and adequate in size for inspection. Participants primary eye-tracking data and verbal responses (Table A1), indicated that although participants were looking for various clues in the images, they mainly focused on the eyes, nose mouth and hair.

Table A1

Eye-tracking and verbal responses of participants.

Subject number and eye-tracking	Fixation behavior profile (eye- tracking recording)	% of accuracy on decision making	% Self-perception of performance	Participants response to the question: how did you scan the pictures?
1	First fixation between the two eyes and most fixations focused only on the eyes and nose. last fixation diverse.	25 % correct	70 % correct	“I looked at the eyes and then at the rest of the face. I looked for places that seemed “inhuman” to me”
2	First fixation between the two eyes and nose, then far-head and clock way across the hair (12–6)	8 % correct	90 % correct	“I examined the person in a preliminary way, then I examined the background and the cover, next I looked at the facial features and looked for imperfections. In addition, I also thought about AI’s bias and does it make sense for an AI engine to create an image of the person”

(continued on next page)

Table A1 (continued)

Subject number and eye-tracking	Fixation behavior profile (eye-tracking recording)	% of accuracy on decision making	% Self-perception of performance	Participants response to the question: how did you scan the pictures?
3	First fixation mainly on far-head and then nose, continuing (with a strong emphasize) to the hair and/or ears and/or neck	75 % correct	70 % correct	"I looked first at the backgrounds and the ears, then I examined the faces themselves if I didn't spot anything that stood out"
4	Variety of fixations with no definition of gaze pattern	25 % correct	60 % correct	"I looked for "flaws" in the photos such as an uneven hairline, marks under the eyes, wrinkles around the eyes when smiling etc."
5	Variety of fixations with no definition of gaze pattern	50 % correct	70 % correct	"I looked for some signs that didn't look realistic"
6	First fixation on the mouth, then the eyes and nose.	25 % correct	60 % correct	"The number of "flaws" in the image compared to an "ideal" appearance"
7	First fixation between the two eyes and nose, then clock way across the head/hair and shirt.	33 % correct	70 % correct	"I looked for imperfections, duplications in the neck, eye color"

An additional indication for the mean value of the fixations process is seen in the "heat maps" below (Fig.A1), presenting a sampling of performances on 4 images (stimuli) by "heat maps". The heat maps present the mean of fixation count: red-the highest number of fixations, yellow -medium, green-lowest.

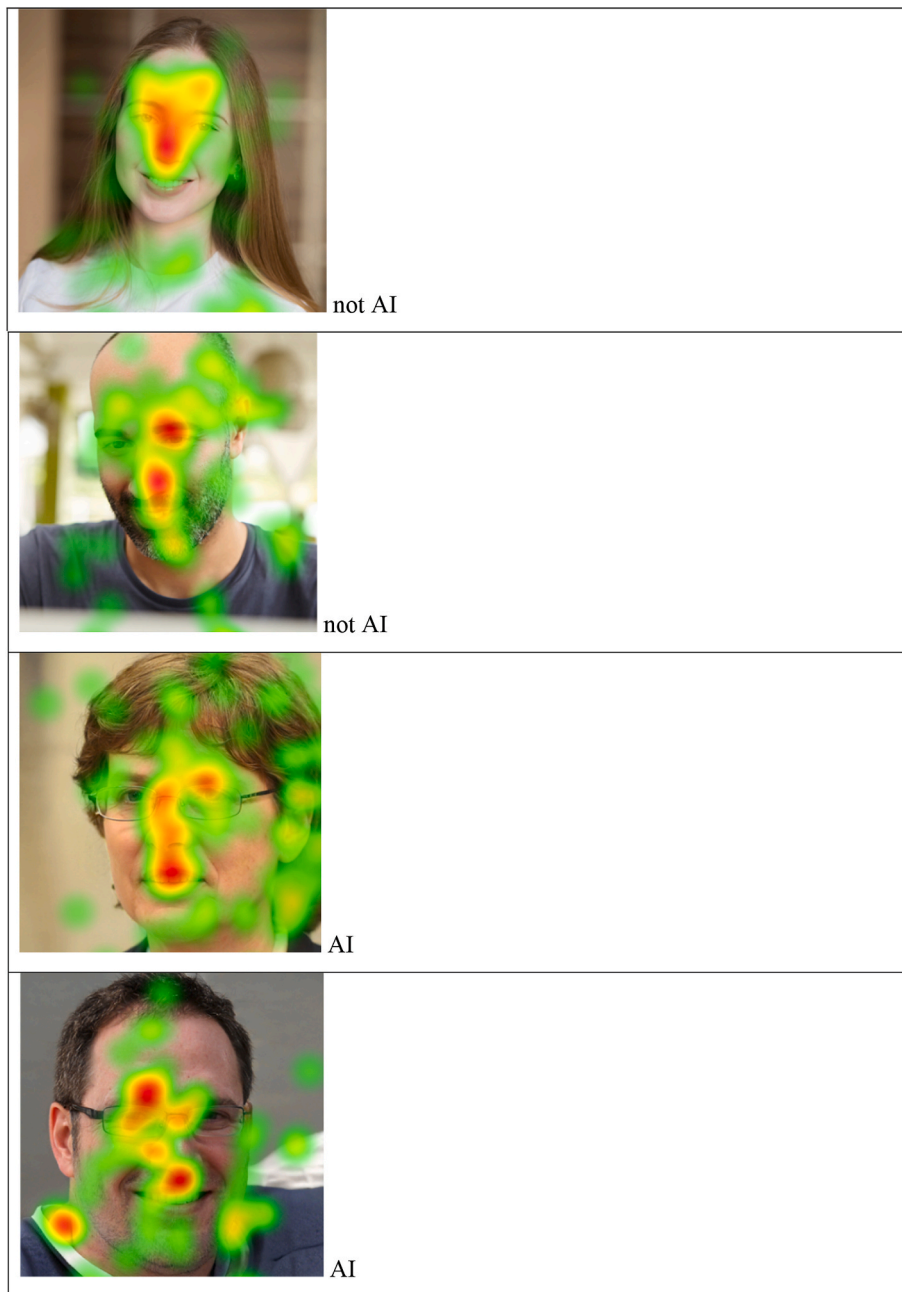


Fig. A1. Heat maps of all participants across 4 images.

Appendix B. Data cleaning

Data cleaning procedure was implemented for each participant, removing fixations that exceeded their individual mean fixation duration by more than four standard deviations. As presented in Fig. B1, on average, 2.4 fixations were excluded per participant, corresponding to approximately 0.9 % of their total fixations.

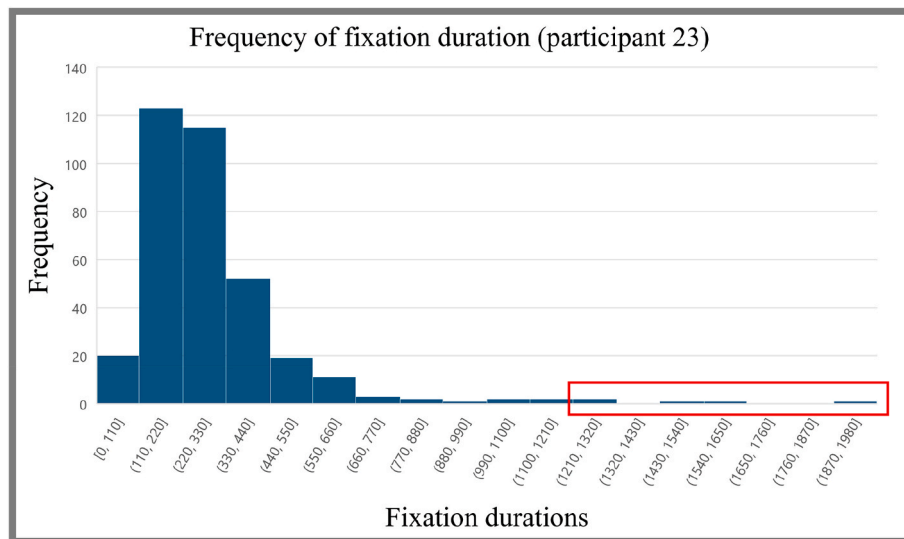


Fig. B1. Histogram depicting the distribution of fixation durations for a sample (participant 23). The mean fixation duration is 300 ms with a standard deviation of 216 ms. Fixations exceeding 1163 ms (mean + 4 SD) were removed, totaling five fixations, which are marked in red.

Appendix C. Source of Fixation Differences

To assure that the significant difference in total fixation duration in background areas did not occur due to an extension of individual total fixation durations or an increase in the number of fixations, we repeated the same analysis. However, this time, instead of examining the total fixation duration in each area of interest, we analyzed the duration of individual fixations within each area of interest. This analysis allows us to determine whether the difference in total fixation duration is due to an increase in fixation length or an increase in the number of fixations.

Results (Figure C1), present the distribution of single fixation durations across all areas of interest is similar, with no significant difference between the experiment and control groups.

For example, when analyzing all fixations (Full image), it was found that there was no significant difference in the woman face image between the experiment and control groups, the average single fixation duration in the experiment group was 288 ± 193 ms, while control group was 301 ± 171 ms (t -test, $p = .17$). Similarly, for the male face image, the average single fixation duration in the control group was 290 ± 148 ms, whereas in the experiment group was 289 ± 180 ms, and in the control group was 290 ± 148 ms (t -test, $p = .90$).

Statistical significance tests conducted for the other AOI's produced similar results, indicating that the distribution of single fixation durations did not differ significantly between the control and experiment groups.

These findings suggest that the difference in total fixation duration stems from a difference in the number of fixations rather than their individual durations. We separately examined whether this difference indeed originated from the number of fixations, and the results support this claim (*data not shown*).

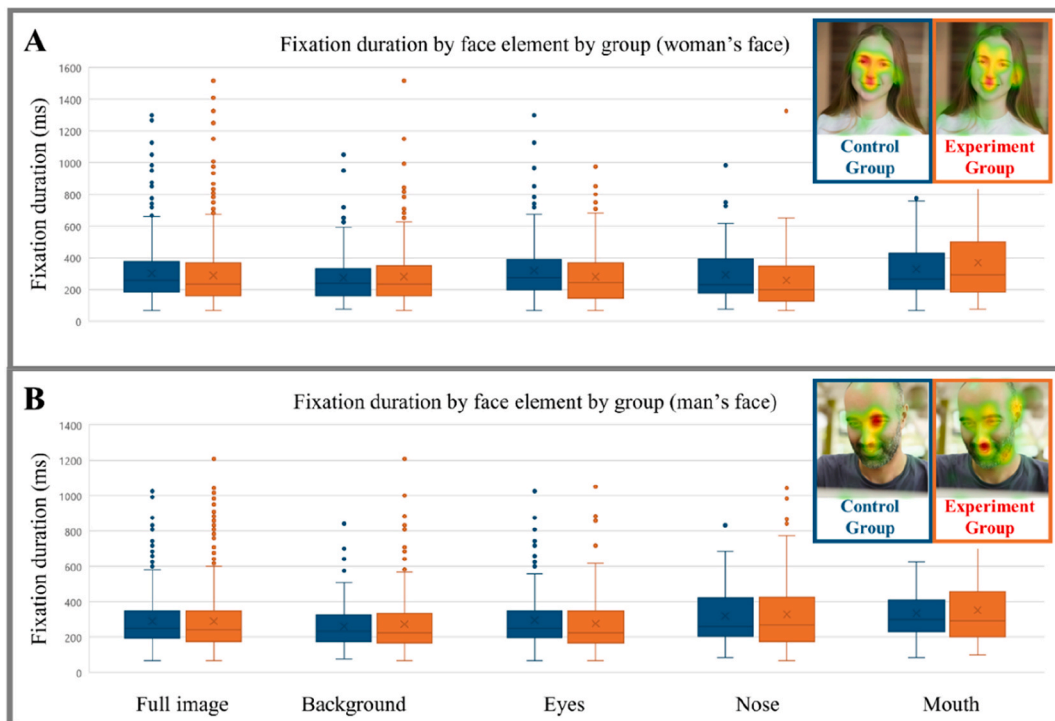


Fig. C1. Distribution of single fixation durations in the experiment and control groups for the woman face image (A) and man face image (B).

Data availability

Data will be made available on request.

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